



COEP TECHNOLOGICAL UNIVERSITY
(COEP TECH)
(Formerly College of Engineering Pune)

T2 Examination

(CT-20014) Data Structures and Algorithms – II

Course: B.Tech , Semester IV

Branch: Computer Engineering

Academic Year: 2023-2024

Max.Marks:20

Duration: 1 Hours

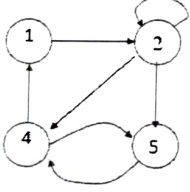
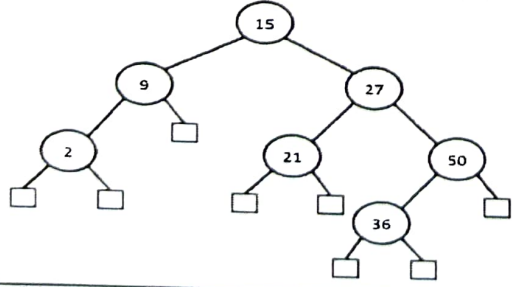
Date: 19/03/2024

Instructions:

Student MIS No.

612203021

1. Figures to the right indicate the full marks.
2. Mobile phones and programmable calculators are strictly prohibited.
3. Writing anything on question paper is not allowed.
4. Exchange/Sharing of stationery, calculator etc. not allowed.
5. Write your MIS Number on Question Paper

			Marks
Q 1	a	How many maximum and minimum nodes will be there for AVL tree of height 3?	2
	b	In given graph, find out in-degree and out-degree of 2 	2
	c	i) What is the advantage of an AVL tree over a BST? ii) Complete graph has _____ number of edges.	2
Q 2	a	i) Show the AVL tree that results after deletion of the integer keys 9 and 27. ii) Explain process to re-balance after deletion of the given keys. iii) Clearly state the imbalanced node, type of imbalance after the operation. 	4
	b	Tree contains the nodes in given order: 8, 2, 4, 3, 7, 6, 5. Show working of Heap sort to get ascending order.	4
Q 3	a	Write a function in C to perform LL rotation in AVL tree.	3
	b	Write a function in C to insert element in a max heap using array.	3

